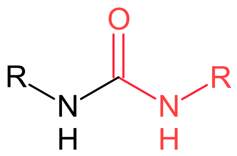
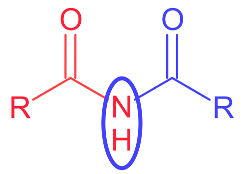
1. RCONH2 can overlap only by N atom but not carbonyl C=O fragment.

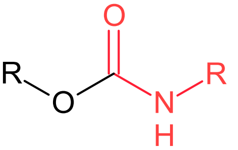


1 amide groups



2 amide groups
connected via NH

2. RC(=O)OR can overlap only by O atom but not carbonyl C=O fragment. Analogous conditions as for 1. + C=O cannot match for RO-C(=O)-OR fragment!
3. Seniority rule: RCONH2 > Ar-CONH2 > RCOOR > Ar-COOR >COOH >Ar-COOH > C=O



SOLUTION for 1. and 2. : In the case of self-overlap, recognize carbonyl C=O group. If C=O is matched twice forbid self-overlap. Otherwaise – allow.

4. Fix Cl-CR2-CR2-Cl and R2CCl2:
- self-overlap allowed via one C-C bond (The same rule for R2CCl2 bond type).
- Cl-CR2-CR2-Cl and R2CCl2 are allowed to match only via one C-C bond.
- Ring and Cl-CR2-CR2-Cl overlap allowed only via two C-C bonds (The same rule for R2CCl2 bond type).

SOLUTION for 4. : If R2CCl2 and Cl-CR2-CR2-Cl are self-overlapped, recognize C-C pair (two C connected by single bond). Allow only for one such pair in two the same bond types. R2CCl2 and Cl-CR2-CR2-Cl (not self) overlap is resolved by SMARTS. The ring and Cl-CR2-CR2-Cl overlap can be resolved by checking how many atoms belong to the same ring. If more than 2 -> exclude matching.

5. Fix Br-CR2-CR2-Br:
- self-overlap allowed via one C-C bond only.
- Ring and Br-CR2-CR2-Br overlap allowed only via two C-C bonds.

SOLUTION for 5. Use some solutions from 4.

6. Fix Ar-NR2:
- If one R = Ar -> assign two Ar-NR2 bond types; for two R = Ar -> match three Ar-NR2.
- For [N+]- Ar4 -> match four Ar-NR2

SOLUTION: If Ar-NR2 is matched check how many carbon atoms are aromatic. The number of matched groups must be equal to the number of aromatic C atoms.

7. Fix Ar-OR:
- If Ar = R, match two Ar-OR bonds

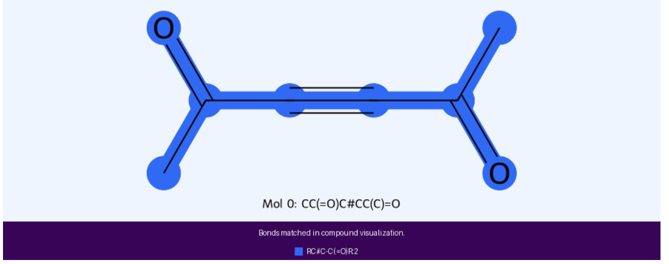
SOLUTION: Analogous as for 6.

8. C=C=C issue:
- Our current logic supporting SMARTS exclude matching of two C=C bond types in C=C=C.

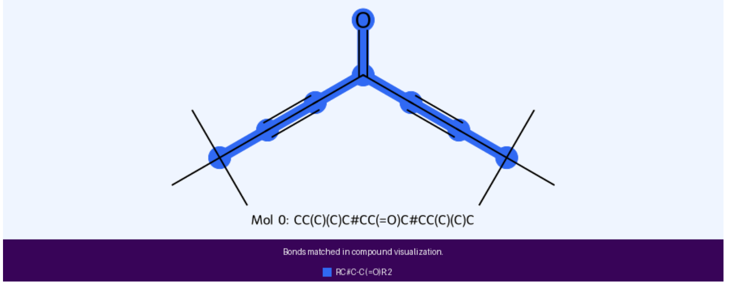
SOLUTION: Add additional condition in the logic. For example use SMARTS that will convey the structural pattern that need to be excluded from the rule. Or if one C atom forms two double bonds with other C atom -> allow for C=C self-overlap.

9. Fix self-overlap of RC#C-C(=O)R. There are two possibilities of of self-overlap:

- via C#C-C fragment.



- via C(=O)C



SOLUTION: Allow for self-overlap via C-C exclusively. Recognize single bonded C atoms (C-C). Only this pair is allowed to overlap. Otherwise -> forbid self-overlap.